

HIRUNA VISHWAMITH

[github/hiruna](#) ◇ [linkedin.com/hiruna](#) ◇ [portfolio/hiruna](#) ◇ [scholar/hiruna](#) ◇ hvishwamith@gmail.com

EDUCATION

University of Moratuwa, Sri Lanka

Jun 2021 - Jul 2025

- BSc. (Hons) in Electronics and Telecommunications Engineering
- Minor in Mathematics
- GPA: 3.75/4.0, Deans List 1st, 2nd, 6th, 8th Semesters

TECHNICAL SKILLS

Hardware & Computer Architecture:	Verilog, RISC-V, FPGA, Cache Coherency, Parallel Computing
Development Tools:	Xilinx Vivado, Intel Quartus, Vitis, Verilator
Programming:	Python, C++, C, Chisel
Machine Learning & AI Acceleration:	PyTorch, TensorFlow, CUDA

EXPERIENCE

FPGA Design Engineer Internship

Apex Compute

Mar 2025 - present

USA, California

Research Scholar - Internship

National University of Singapore

Dec 2023 - May 2024

Singapore, Clementi

- ❖ Designed and implemented 3DRA CGRA (Coarse-Grained Reconfigurable Architecture) on U250 FPGA, achieving a 650MHz operating frequency
- ❖ Optimized data communication for a 100Gbps TCP/IP Offload Engine.

Project Assistant [?]

University of Moratuwa

Jan 2023 - Apr 2023

Sri Lanka, Moratuwa

- ❖ Contributed to developing an Intelligent Camera with Real-Time Video Analytics System
- ❖ Developed a License Plate Recognizing web app using YOLOv5

[\[web app\]](#)

PUBLICATIONS

- T. Coskun, **H. Vishwamith**, M. Isik and I. Can Dikmen, "Hardware-Accelerated Transformer Framework for Real-Time Battery SoH Estimation," IEEE High Performance Extreme Computing Conference (IEEE HPEC), 2025.
- T. Coskun, **H. Vishwamith**, M. Isik and I. Can Dikmen, "Exploring Neuromorphic Computing with Loihi-2 for High-Performance CFD Simulations," IEEE High Performance Extreme Computing Conference (IEEE HPEC), 2025.
- **H. Vishwamith**, M. Isik and I. Can Dikmen, "HPCNeuroNet: Advancing Neuromorphic Audio Signal Processing with Transformer-Enhanced Spiking Neural Networks," Interdisciplinary Conference on Electrics and Computer (INTCEC), 2024. [Link to : [Paper](#)]
- M. Isik, **H. Vishwamith**, Y. Sur and I. Can Dikmen, "NEUROSEC: FPGA-Based Neuromorphic Audio Security," International Symposium on Applied Reconfigurable Computing (ISARC), 2024. [Link to : [Paper](#)]
- M. Isik, **H. Vishwamith**, J. Naoukin and I. Can Dikmen, "HPCNeuroNet: A Neuromorphic Approach Merging SNN Temporal Dynamics with Transformer Attention for FPGA-based Particle Physics," Conference on Computing in High Energy and Nuclear physics (CHEP), 2024. [Link to : [Paper](#)]

FINAL YEAR PROJECT

Cache coherent, Out of Order, Multi-core RISC-V CPU [?]

- Developed a parametric multi-core RISC-V emulator capable of running bare-metal multi-core programs and booting Linux in multi-core mode.
- Developed an Instruction Fetch Unit (IFU) with a TAGE branch predictor, which is capable of achieving the desired performance goals on an FPGA.
- Developed a multi-threading benchmark suite for benchmarking and testing multi-core RISC-V CPU
- Implemented a Cache Coherent, Out of Order, Multi-core RISC-V CPU on FPGA capable of booting Linux in multi-core mode.

COMPETITIONS

ACM/IEEE TinyML Design Contest @ ICCAD 2023

Sep 2023 - Nov 2023

- ❖ Implementing deep learning algorithm that can automatically discriminate VAs (i.e., Ventricular Fibrillation and Ventricular Tachycardia) [\[leaderboard\]](#)
- ❖ Deployed model on STM32F303K8T6 (ARM Cortex M4).
- ❖ Optimized inference latency to achieve **2nd place in latency score** with latency 0.52ms

IEEE Signal Processing Society VIP Cup 2023 [?]

Aug 2023 - Sep 2023

- ❖ Developed a ML model using SIMCLR for VIP Cup 23 - IEEE Signal Processing Society [\[leaderboard\]](#)
- ❖ The ML model was capable of identifying 6 Biomarkers of OCT images using clinical data
- ❖ I was able to achieve F1 score of 70% in the final phase

PROJECTS

SIMD Processor based SoC on FPGA

Oct 2024 - Dec 2024

- ❖ Developed an SoC integrated with a hardware accelerator on the Xilinx Zybo Z7 FPGA board.
- ❖ Algorithms have been developed and tested for matrix multiplication, dot product, and other operations.
- ❖ Parameterized architecture designed to generate configurations for any number of Processing Engines (PE); architectures with 2 to 32 PE counts have been tested.
- ❖ Achieved a 200 MHz operating frequency on FPGA by efficiently utilizing DSP blocks and BRAMs.

System bus on FPGA

Oct 2024 - Dec 2024

- ❖ Developed a system bus with 2 masters and 3 slaves on the DE0 FPGA board.
- ❖ The architecture includes a priority-based arbiter that supports split transactions.
- ❖ Implemented and tested bus bridges that support connecting two system buses implemented on two different FPGAs.

Three Stage RISC-V Processor on FPGA

Jul 2023 - Jan 2024

- ❖ Developed a pipelined RISC-V processor with three stages on the Xilinx Zybo Z7 FPGA board.
- ❖ As part of the development cycle, separate processors with single-cycle, multi-cycle, and pipelined architectures were implemented.
- ❖ Verified functionality using testbenches and waveform diagrams.